

HEALTHCONNECT CLINIC — WEEK 5

PROJECT SUMMARY — DATA ANALYTICS TRACK

1. Planned Work

Week 5 was planned as the transition from Week 4 planning and KPI definition into practical analytical execution: complete data preparation, explore meaningful relationships, calculate selected KPIs, identify findings, formulate recommendations and document limitations.

2. Completed Work

The Week 4 processed dataset was carried forward into Week 5. Data-quality checks and remaining missing-value treatment were completed, followed by exploratory analysis of appointment outcomes, appointment type, reminder behaviour, previous no-show history, distance to clinic, waiting time and numerical relationships.

3. KPI Results

KPI	RESULT / RELEVANCE
ATTENDANCE RATE	46.28%. Establishes the baseline proportion of successfully attended appointments.
NO-SHOW RATE	48.46%. Directly measures the project's core missed-appointment problem.
REMINDER EFFECTIVENESS	Compares attendance when a reminder was sent with attendance when no reminder was sent.
ATTENDANCE RATE BY REMINDER CHANNEL	Compares attendance performance across reminder channels for more actionable reminder analysis.

4. Key Findings

1. No-shows are the largest appointment outcome: 48.46% no-shows versus 46.28% attended and 5.26% cancelled.

This confirms that missed appointments are a substantial operational issue.

2. Follow-up appointments have the highest observed no-show rate at 51.23%, indicating that appointment type maybe relevant when considering attendance-support strategies.
3. Reminder behaviour shows a clear pattern: no-reminder appointments have the strongest no-show pattern, followed by WhatsApp, email and SMS in the observed ranking. Reminder strategy therefore warrants further investigation.
4. Previous no-show history appears relevant. Patients with five previous no-shows all missed their current appointment, and no-show percentage showed a slight increase as previous no-shows increased.

5. Distance to clinic does not appear to meaningfully affect no-show behaviour. The median distance among attendees was only slightly lower than among cancelled and no-show appointments.
6. Waiting time does not appear to meaningfully affect no-show behaviour, although cancelled appointments had a slightly lower median waiting time.
7. No meaningful correlation was observed among the numerical variables examined, suggesting that simple linear relationships alone do not explain the attendance problem.

5. Business Recommendations

- Prioritise patients with previous no-show history for targeted attendance-support interventions.
- Review and strengthen reminder coverage, particularly for appointments currently receiving no reminder.
- Compare reminder channels using attendance outcomes before prioritising one channel.
- Examine follow-up appointments more closely and assess whether targeted reminder or follow-up actions are appropriate.
- Do not treat distance or waiting time as primary explanations for no-shows based on the current evidence.

6. Limitations

The analysis identifies associations rather than causal relationships. The dataset may not capture every behavioural or operational factor affecting attendance. Missing-value treatment can affect a small portion of the analysis. The Week 5 results are an initial evidence base and should be strengthened through later modelling, validation and intervention testing.

7. Decisions and Changes from Week 4

The project progressed from Week 4 KPI definition and analytical planning to actual KPI calculation and exploratory evidence. The analysis retained the Week 4 focus on attendance and no-shows while narrowing attention toward reminder behaviour, previous no-show history and appointment characteristics.

8. Collaboration and Remaining Work

The Data Analytics findings provide useful input for Data Science, particularly around previous no-shows, reminder variables and appointment type. Remaining work includes further validation, cross-track integration and progression toward the next project stage.

9. Week 6 Focus

Build on the strongest Week 5 findings, deepen cross-track integration and translate the evidence into the next stage of the HealthConnect solution while maintaining clear documentation of assumptions and limitations.